Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An impactor has 3 times the original mass at the same speed and stopping time. Compare momentum and average stopping-force size.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

Worked reasoning: A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two equal-mass carts move at speeds 2 and 4 m/s. Predict how their momenta and kinetic energies compare before calculating.



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

The lab carts in question 3 each have mass 2 kg. One moves at 2 m/s and the other at 4 m/s. Calculate both kinetic energies. Draw bars to test your prediction about the effect of doubling speed.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

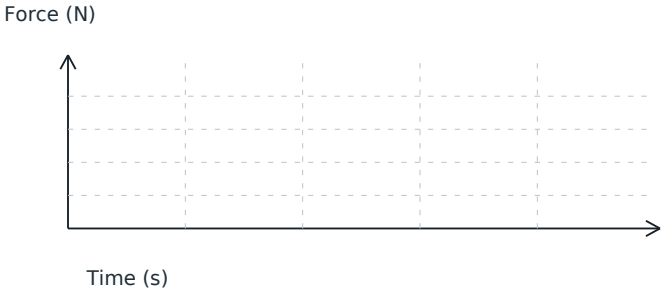
05 Explain your reasoning

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 24 N net force acts right for 3 s on a 4 kg cart initially at rest. Find impulse and final velocity.
Show the signed impulse on the force-time model.



Marzano check - circle: 1 2 3

07 Exit check

For the same mass, speed triples. By what factor does kinetic energy change? Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find two carts with equal mass but speeds in a 1: 2 ratio. Compare momentum and energy multipliers. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3